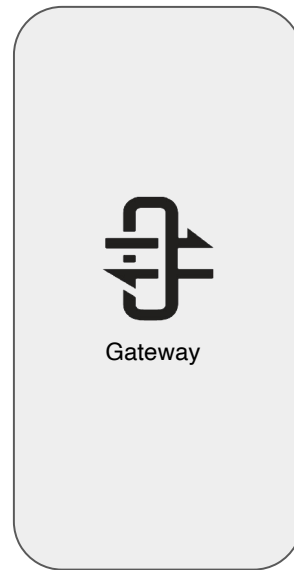


Tyk Product Components: Gateway & Redis

Exploring the role of each component in the Tyk Ecosystem

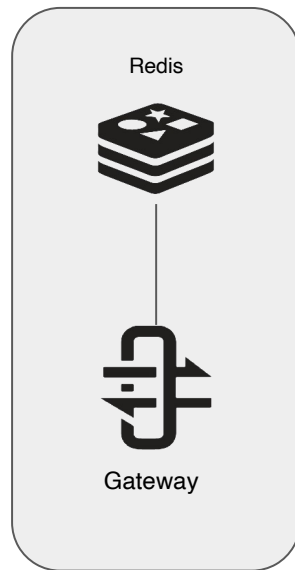
Tyk Gateway

- Real Time Traffic Enforcement
- Handles all inbound API traffic
- Authentication & Authorization
- Rate Limiting and Quotas
- Enforces Policies
- Observability



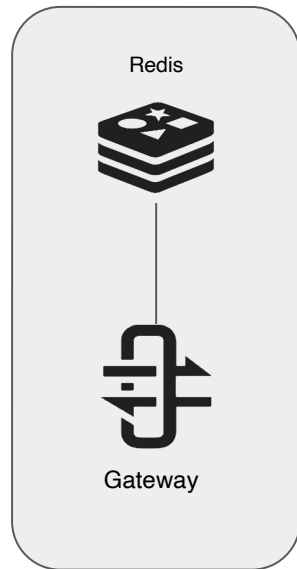
Tyk & Redis

- Fast in-memory performance
 - Enables real-time request validation
 - Minimise latency for high-throughput APIs
- Stores:
 - Rate limit counters
 - Session tokens
 - API key states
- Shared State across gateways
 - Synchronizes usage data across nodes
- Pub/Sub Broker
 - Cluster-wide notifications



Redis High Availability in Tyk

- Redis is a single point of failure
 - Tyk Gateway depends on Redis for live traffic processing
 - Redis unavailability = Gateway downtime
- HA with Redis Sentinel
 - Monitor & Automatic Failover
- Redis Clustering
 - Distribute data across multiple shards



Redis Cluster: High-Availability

- Distributed implementation of Redis for horizontal scaling and high availability.
- It automatically partitions data across multiple nodes and supports failover.
- Redis Cluster uses a sharded data set:
 - Horizontal scaling option
- Primary-Replica Model: Each primary node has at least one replica.
- Minimum of 3 Master instances required:
 - Master majority required for failover
- Gossip Protocol: Nodes communicate to detect failures and trigger failovers.

